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1. RESEARCH INTERESTS

- Generative AI-based data augmentation for construction computer vision
- Image quality assessment for construction object detection
- Computer vision and machine learning for construction safety management
- Large language models and retrieval-augmented generation for construction supervision

2. EDUCATION

M.S. Architectural Engineering (Construction Management) ..... 03/2025 - 08/2026
- Pusan National University, Busan, South Korea (Advisor: JungHo Jeon)
- Thesis: Improving Construction Object Detection Using Generative AI Image Quality Assessment

B.S. Architectural Engineering ..... 03/2021 - 02/2025
- Pusan National University, Busan, South Korea (Advisor: JungHo Jeon)

3. ACADEMIC EXPERIENCE

Teaching and Research Support (TA/TB) ..... 09/2025 - 08/2026
- Department of Architectural Engineering, Pusan National University
- Main Tasks: Teaching and research support

4. RESEARCH EXPERIENCE

4.1 Research Projects

Construction Safety Management Agents Using Human Foundation Models ..... 09/2026 - 02/2027
- KRW 90,153,000; national R&D
- Role: Participating Researcher, Pusan National University

Safety Management Platform Using Generative AI Data Augmentation ..... 03/2025 - 02/2030
- KRW 1,237,335,000; National Research Foundation of Korea
- Role: Participating Researcher, Pusan National University

BIM-Based Decision Support for Green Building Planning Across Design Stages ..... 06/2024 - 02/2026
- KRW 1,344,502,000; supported by the National Research Foundation of Korea.
- Role: Participating Researcher, Pusan National University

Checklists for Finishing Work Based on Defect Risk by Building Type ..... 06/2024 - 08/2024
- KRW 54,788,000; national R&D
- Role: Participating Researcher, Pusan National University

Intelligent Robotic System for Grinding High-Rise Concrete Walls ..... 05/2024 - 02/2025
- KRW 174,144,000; national R&D
- Role: Participating Researcher, Pusan National University

4.2 Peer-Reviewed Journal Articles (2 published and 1 under review)

[03] **Lee, Y.**, Lee, J., Baek F., Kim H., Kim J., & Jeon J. (2026). "Synthetic Data Filtering for Enhanced Construction Object Detection via Image Quality Assessment". Developments in the Built Environment (*Under- review*)

[02] Jung, D., **Lee, Y.**, Jeong, K., Lee, J., Kim, J., Park, H., & Jeon, J. (2025). "Improvement of Construction Workers' Drowsiness Detection and Classification via Text-to-Image Augmentation and Computer Vision". Sustainability, 17(20), 9158. https://doi.org/10.3390/su17209158

[01] **Lee, Y.**, Kang, G., Kim, J., Yoon, S., & Jeon, J. (2025). "Generative AI-driven data augmentation for enhanced construction hazard detection". Automation in Construction, 177, 106317. https://doi.org/10.1016/j.autcon.2025.106317

4.3 International Conference Papers (1 published and 1 forthcoming)

[02] **Lee, Y.**, Yoon S., Kim N., & Jeon J. (2026). "Quality-centered Generative Image Augmentation Approach for Construction Safety Object Detection". International Conference on Computing in Civil Engineering 2026. (Presented; proceedings Forthcoming.)

[01] Jo, W., **Lee, Y.**, Jung, D., Park, H., & Jeon, J. (2024). "Computer Vision-based Construction Hazard Detection via Data Augmentation Approach using Generative-AI". Proceedings of the International Conference on Construction Engineering and Project Management, 791-798. Korea Institute of Construction Engineering and Management. https://doi.org/10.6106/ICCEPM.2024.0791

4.4 Domestic Conference Papers (6 published)

[06] **Lee, Y.**, Jo, W., Lim, G., Kim, G., Park, H., & Jeon, J. (Nov. 2025). Assessing the Potential of Image Quality Assessment to Improve Construction Site Object Detection. Proceedings of the 2025 Annual Conference of the Korea Institute of Construction Engineering and Management, 29-30. (Best Paper Award.)

[05] Seong, J., Jeong, K., **Lee, Y.**, Kang, G., Park, H., & Jeon, J. (Nov. 2025). Multiclass Classification of Construction Workers' Stress States Using rPPG. Proceedings of the 2025 Fall Conference of the Korea Institute of Building Construction, 25(2), 71-72. (Best Paper Award.)

[04] Jo, W., **Lee, Y.**, Jeong, K., Park, G., Choi, Y., Kim, G., Park, H., & Jeon, J. (Oct. 2025). Comparing Construction Duration Prediction Performance Using Machine Learning and Generative AI. Proceedings of the Architectural Institute of Korea Conference, 45(2), 766-769. (Best Paper Award.)

[03] **Lee, Y.**, Kang, G., Jo, W., Seong, J., Park, H., & Jeon, J. (Apr. 2025). Development of an AI-Based Generative Image Quality Assessment Framework for Construction Safety Management. Proceedings of the Architectural Institute of Korea Conference, 45(1), 669-670.

[02] **Lee, Y.**, Kang, G., Jo, W., Cho, S., & Jeon, J. (Dec. 2024). A Framework for Improving Safety Helmet Detection at Construction Sites Through Generative AI Data Augmentation. Proceedings of the Busan, Ulsan, and Gyeongnam Chapter Conference of the Architectural Institute of Korea, 357-358.

[01] Kang, G., **Lee, Y.**, Jo, W., Cho, S., Yoon, S., & Jeon, J. (Oct. 2024). Generative AI Data Augmentation for Improved Computer Vision Hazard Identification. Proceedings of the Architectural Institute of Korea Conference, 44(2), 1184-1187. (Best Paper Award.)

5. HONORS AND AWARDS

Best Paper Award, Korea Institute of Construction Engineering and Management 11/2025
Outstanding Academic Award, Korea Institute of Building Construction 11/2025
Outstanding Presentation Paper Award, Architectural Institute of Korea 10/2025

Top Prize, Construction Scheduling Competition, KICEM 11/2024
Best Paper Award, Architectural Institute of Korea 10/2024
Encouragement Award, Cost Management Competition, KICEM 11/2023

6. COMPETENCES

Programming Languages: MATLAB, Python, Program R, and C++
Software & Frameworks: AutoCAD, Revit, P6 MS project, SketchUp, PyTorch, YOLO, and Stable Diffusion
Languages: Korean (Native), English
Other: MS Office

7. REFERENCES

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